

KCS

KCS stands for Knowledge-Centered Service, a methodology that aims to improve service delivery and knowledge management in service organizations. Some of the main features and benefits of KCS are:

- KCS is based on the principle of “capture, structure, reuse, and improve” knowledge as part of the service process.
- KCS enables service agents to create and update knowledge articles in real time, based on the issues and solutions they encounter during service interactions.
- KCS empowers service agents to search and reuse existing knowledge articles to solve problems faster and more consistently, reducing duplication and rework.
- KCS fosters a culture of collaboration and feedback among service agents, who can rate, comment, and flag knowledge articles for quality and relevance.
- KCS leverages analytics and metrics to measure and improve the performance and value of the knowledge base and the service organization.
- KCS aligns with the business goals and strategies of the service organization, such as increasing customer satisfaction, reducing costs, and enhancing innovation.

Unit 1 - Brief history of microbiology, Types of microorganisms, Basic idea of domain bacteria, proteobacteria, non proteobacteria Gram –ve and Gram +ve bacteria, lichens, algae, protozoa, helminthes, viral structures, viral multiplication, Role of microorganisms in the production of industrial chemicals and pharmaceuticals

Brief history of microbiology

- Microbiology is the study of microscopic organisms, such as bacteria, viruses, fungi, protozoa, and algae.
- Microbiology essentially began with the development of the microscope in the 17th century.
- Antonie van Leeuwenhoek (1632–1723) was one of the first people to observe microorganisms, using a microscope of his own design, and made one of the most important contributions to biology .
- Robert Hooke (1635–1703) was the first to use a microscope to observe living things, such as fungi, algae, and cork cells .
- Ferdinand Cohn (1828–1898) was a botanist who founded the field of bac-

teriology and described several bacteria, including *Bacillus* and *Beggiatoa*.

- Louis Pasteur (1822–1895) was a chemist who demonstrated the role of microorganisms in fermentation, spoilage, and disease, and developed the process of pasteurization and the germ theory of disease.
- Robert Koch (1843–1910) was a physician who established the criteria for proving the causal relationship between a microbe and a disease (Koch's postulates) and identified the causative agents of anthrax, tuberculosis, and cholera.
- Other notable figures in the history of microbiology include Joseph Lister, Alexander Fleming, Martinus Beijerinck, Sergei Winogradsky, Paul Ehrlich, Elie Metchnikoff, and Carl Woese.

Types of microorganisms

- Microorganisms are classified into different domains, kingdoms, phyla, classes, orders, families, genera, and species based on their genetic and phenotypic characteristics.
- The three domains of life are Bacteria, Archaea, and Eukarya.
- Bacteria and Archaea are prokaryotes, which means they lack a nucleus and other membrane-bound organelles. They are usually unicellular and have diverse shapes, sizes, metabolisms, and habitats.
- Eukarya are eukaryotes, which means they have a nucleus and other membrane-bound organelles. They can be unicellular or multicellular and include animals, plants, fungi, and protists.
- Fungi are eukaryotic organisms that obtain nutrients by absorbing organic matter from their environment. They include molds, yeasts, mushrooms, and lichens.
- Algae are eukaryotic organisms that perform photosynthesis and produce oxygen. They include seaweeds, diatoms, dinoflagellates, and green algae.
- Protozoa are eukaryotic organisms that are usually motile and feed on other microorganisms or organic matter. They include amoebas, flagellates, ciliates, and sporozoans.
- Helminthes are eukaryotic organisms that are multicellular worms, such as roundworms, tapeworms, and flukes. They are parasites that infect animals and humans.
- Viruses are acellular entities that consist of a nucleic acid (DNA or RNA) surrounded by a protein coat. They are obligate intracellular parasites that can only replicate inside a living host cell.

Basic idea of domain bacteria, proteobacteria, non proteobacteria Gram –ve and Gram +ve bacteria

- Bacteria are the most abundant and diverse domain of life. They are found in almost every environment on Earth and play important roles in biogeochemical cycles, symbiosis, and human health and disease.

- Bacteria are classified into different phyla based on their genetic and phenotypic characteristics, such as cell wall structure, metabolism, morphology, and ecology.
- One of the most common methods to classify bacteria is based on their reaction to the Gram stain, which differentiates bacteria into Gram-positive and Gram-negative based on the presence or absence of a thick peptidoglycan layer in their cell wall.
- Gram-positive bacteria have a thick pe

Brief history of microbiology

- Microbiology is the scientific study of microorganisms, which are usually too small to be seen with the naked eye.
- Microbiology has its origins in the 16th century, when some people observed the existence of invisible living things that caused diseases or fermentation.
- The development of the microscope in the 17th century enabled the direct observation of microorganisms by scientists such as Robert Hooke and Anton van Leeuwenhoek.
- Robert Hooke (1635-1703) was an English scientist who coined the term “cell” after observing cork and plant tissues under a microscope. He also observed strands of fungi and microfossils.
- Anton van Leeuwenhoek (1632-1723) was a Dutch draper and amateur microscopist who made his own lenses and microscopes. He was the first to observe and describe bacteria, protozoa, algae, and other microorganisms, which he called “animalcules” or “little animals” .
- In the 18th and 19th centuries, microbiology became a more established and diverse field, with contributions from many scientists such as Ferdinand Cohn, Louis Pasteur, Robert Koch, Joseph Lister, and others.
- Ferdinand Cohn (1828-1898) was a German botanist and microbiologist who founded the discipline of bacteriology. He classified bacteria into four groups based on their shape and mode of reproduction. He also discovered bacterial spores and endospores.
- Louis Pasteur (1822-1895) was a French chemist and microbiologist who demonstrated the role of microorganisms in fermentation, spoilage, and disease. He also developed the germ theory of disease, the process of pasteurization, and several vaccines against anthrax, rabies, and other diseases.
- Robert Koch (1843-1910) was a German physician and microbiologist who established the criteria for proving the causal relationship between a microbe and a disease, known as Koch’s postulates. He also identified the causative agents of anthrax, tuberculosis, cholera, and other diseases.
- Joseph Lister (1827-1912) was a British surgeon and microbiologist who applied the germ theory of disease to surgery and introduced the use of antiseptics to prevent wound infections. He also developed methods for

sterilizing surgical instruments and dressings.

- In the 20th and 21st centuries, microbiology has expanded to include many subdisciplines and applications, such as virology, immunology, molecular biology, biotechnology, genomics, metagenomics, microbiome, and others. Microbiology has also contributed to the fields of medicine, agriculture, industry, and environmental science.

Types of microorganisms

Microorganisms are microscopic organisms that exist as unicellular, multicellular, or cell clusters. They can be beneficial or harmful to life, and they can be divided into six major types :

- **Bacteria:** These are prokaryotic cells, meaning they lack a nucleus and other membrane-bound organelles. They have a cell wall made of peptidoglycan, and they can be classified by their shape, Gram stain, oxygen requirement, and metabolic diversity. Some examples of bacteria are *Escherichia coli*, *Staphylococcus aureus*, and *Streptococcus pneumoniae*.
- **Archaea:** These are also prokaryotic cells, but they differ from bacteria in their cell wall composition, membrane lipids, and ribosomal RNA sequences. They are often found in extreme environments, such as hot springs, salt lakes, and deep-sea vents. Some examples of archaea are *Methanobacterium*, *Halobacterium*, and *Thermococcus*.
- **Protozoa:** These are eukaryotic cells, meaning they have a nucleus and other membrane-bound organelles. They are usually motile and heterotrophic, meaning they obtain their nutrients by ingesting other organisms or organic matter. They can be classified by their mode of locomotion, such as flagella, cilia, or pseudopodia. Some examples of protozoa are *Paramecium*, *Amoeba*, and *Plasmodium*.
- **Algae:** These are also eukaryotic cells, but they are photosynthetic, meaning they use light energy to synthesize organic molecules. They have a cell wall made of cellulose, and they can be unicellular or multicellular. They can be classified by their pigments, such as chlorophyll, carotenoids, or phycobilins. Some examples of algae are *Chlamydomonas*, *Spirogyra*, and *Ulva*.
- **Fungi:** These are also eukaryotic cells, but they are saprophytic, meaning they decompose organic matter. They have a cell wall made of chitin, and they can be unicellular or multicellular. They can be classified by their mode of reproduction, such as sexual or asexual spores. Some examples of fungi are *Saccharomyces*, *Penicillium*, and *Aspergillus*.
- **Viruses:** These are acellular entities, meaning they are not composed of cells. They consist of a nucleic acid (DNA or RNA) enclosed in a protein coat, and sometimes a lipid envelope. They are obligate intracellular parasites, meaning they can only replicate inside a host cell. They can be classified by their shape, size, genome type, and host range. Some

examples of viruses are Influenza, Herpes, and HIV.

Basic idea of domain bacteria

- Domain bacteria is one of the three domains of life, along with domain archaea and domain eukarya.
- Domain bacteria includes the prokaryotes people encounter on an everyday basis. Prokaryotes are organisms that lack a membrane-bound nucleus and other organelles.
- Most bacterial species are heterotrophic; that is, they acquire their food from organic matter. The largest number of bacteria are saprobic, meaning that they feed on dead or decaying organic matter. A few bacterial species are parasitic; these bacteria live within host organisms and cause disease.
- Domain bacteria can be classified into five main phyla: proteobacteria, cyanobacteria, firmicutes, actinobacteria, and spirochaetes.
- Proteobacteria are the largest and most diverse group of bacteria. They include many common pathogens, such as *E. coli*, *Salmonella*, and *H. pylori*, as well as some beneficial bacteria, such as *Rhizobium*, which fix nitrogen in the soil.
- Cyanobacteria are also known as blue-green algae. They are capable of photosynthesis and produce oxygen. They are important for the evolution of life on Earth, as they are believed to be the first organisms to perform oxygenic photosynthesis.
- Firmicutes are mostly gram-positive bacteria, meaning that they have a thick cell wall that retains a purple stain in a laboratory test. They include some of the most common bacteria in the human body, such as *Lactobacillus*, which helps digest food, and *Staphylococcus*, which can cause infections.
- Actinobacteria are also mostly gram-positive bacteria, but they have a high G+C content in their DNA, meaning that they have more guanine and cytosine nucleotides than adenine and thymine. They include some of the most important soil bacteria, such as *Streptomyces*, which produce antibiotics, and *Mycobacterium*, which cause tuberculosis and leprosy.
- Spirochaetes are bacteria that have a spiral shape and a flexible cell wall. They can move by rotating their flagella, which are located inside the cell wall. They include some of the most notorious pathogens, such as *Treponema*, which causes syphilis, and *Borrelia*, which causes Lyme disease.

Proteobacteria

- Proteobacteria is a phylum of gram-negative bacteria discovered by Carl Woese in the 1980s based on nucleotide sequence homology.
- Proteobacteria are further classified into the classes alpha-, beta-, gamma-, delta- and epsilonproteobacteria, each class having separate orders, fami-

lies, genera, and species .

- Proteobacteria include a wide variety of pathogenic and free-living bacteria, such as *Escherichia*, *Salmonella*, *Vibrio*, *Yersinia*, *Legionella*, and many others .
- Proteobacteria also include many of the bacteria responsible for nitrogen fixation, a process that converts atmospheric nitrogen into ammonia, which can be used by plants and other organisms .
- Proteobacteria are characterized by their diverse metabolic pathways, such as aerobic and anaerobic respiration, fermentation, photosynthesis, and chemolithotrophy.
- Proteobacteria are also capable of living in low-nutrient environments such as deep oceanic sediments, glacial ice, or deep undersurface soil.
- Proteobacteria are important for the production of industrial chemicals and pharmaceuticals, such as antibiotics, bioplastics, biofuels, and bioremediation agents.

Nonproteobacteria Gram-Negative Bacteria

- Nonproteobacteria are a diverse group of bacteria that do not belong to the phylum Proteobacteria, which contains the majority of gram-negative bacteria.
- Nonproteobacteria can be either gram-negative or gram-positive, depending on the presence or absence of an outer membrane in their cell wall.
- Nonproteobacteria can have various metabolic modes, such as photoautotrophy, chemoheterotrophy, or chemoautotrophy.
- Some examples of nonproteobacteria are:
 - **Chlamydia**: These are obligate intracellular parasites that infect animals and humans, causing diseases such as trachoma, chlamydia, and lymphogranuloma venereum. They have a unique developmental cycle that involves two forms: elementary bodies and reticulate bodies. They lack peptidoglycan in their cell wall and depend on their host for ATP and other nutrients.
 - **Spirochetes**: These are motile, spiral-shaped bacteria with a long, narrow body. They have a unique structure called the axial filament, which is a bundle of flagella that runs between the inner and outer membranes of the cell wall. They can move by rotating their body like a corkscrew. They are difficult or impossible to culture in the laboratory. Some spirochetes are pathogens, such as *Treponema pallidum* (causes syphilis), *Borrelia burgdorferi* (causes Lyme disease), and *Leptospira* (causes leptospirosis) .
 - **CFB group**: This is a phylum of bacteria that includes the genera *Cytophaga*, *Fusobacterium*, and *Bacteroides*. They share some similarities in their DNA sequences, but they are phylogenetically di-

verse. They are mostly anaerobic and inhabit various environments, such as soil, water, and the human gut. Cytophaga are rod-shaped bacteria that can degrade cellulose and other complex carbohydrates. Fusobacterium are spindle-shaped bacteria that can cause infections in the mouth, throat, and lungs. Bacteroides are the most abundant bacteria in the human colon and play a role in digestion and immunity .

- **Planctomycetes:** These are aquatic bacteria that have some unusual features, such as a lack of peptidoglycan in their cell wall, a compartmentalized cytoplasm, and the ability to produce membrane-bound vesicles. Some planctomycetes can perform anammox, which is a process of anaerobic ammonia oxidation that produces nitrogen gas. This process is important for the global nitrogen cycle.
- **Phototrophic bacteria:** These are bacteria that can use light as an energy source to synthesize organic compounds. They can be either oxygenic or anoxygenic, depending on whether they produce oxygen or not during photosynthesis. Oxygenic phototrophic bacteria include cyanobacteria, which are the most abundant and diverse group of photosynthetic prokaryotes. They have chlorophyll a and phycobilins as pigments and can fix nitrogen in specialized cells called heterocysts. Anoxygenic phototrophic bacteria include purple bacteria, green bacteria, and heliobacteria, which have different types of pigments and electron donors for photosynthesis .

Gram-positive bacteria

- Gram-positive bacteria are bacteria that have a thick cell wall composed of peptidoglycan and teichoic acids .
- Gram-positive bacteria can be distinguished from gram-negative bacteria by the Gram stain test, which involves a chemical dye that stains the cell wall purple or blue .
- Gram-positive bacteria include many important pathogens, such as Staphylococcus, Streptococcus, Bacillus, Clostridium, Corynebacterium, Listeria, and Mycobacterium .
- Gram-positive bacteria can cause various infections, such as skin infections, pneumonia, meningitis, anthrax, tetanus, diphtheria, tuberculosis, and leprosy .
- Gram-positive bacteria can also produce useful substances, such as antibiotics, enzymes, vitamins, and probiotics.
- Gram-positive bacteria can undergo bacterial transformation, which is the uptake and incorporation of DNA from the environment.
- Gram-positive bacteria are classified into two major groups based on their cell wall structure: Firmicutes and Actinobacteria.
- Firmicutes are mostly rod-shaped or spherical bacteria that can form endospores, which are resistant structures that allow them to survive harsh

conditions.

- Actinobacteria are mostly filamentous or branched bacteria that can produce secondary metabolites, such as antibiotics and pigments.

Lichens

- Lichens are **composite organisms** that consist of a **symbiotic association** of algae (usually green) or cyanobacteria and fungi (mostly ascomycetes and basidiomycetes).
- The dominant partner is the **fungus**, which gives the lichen the majority of its characteristics, from its **thallus shape** to its **fruiting bodies**. The alga can be either a green alga or a blue-green alga, otherwise known as **cyanobacteria**.
- Lichens are found **worldwide** and occur in a variety of **environmental conditions**. They are **pioneer species**, among the first living things to grow on bare rock or areas denuded of life by a disaster.
- Lichens have different **growth forms**, such as crustose (crusty), foliose (leafy), fruticose (shrubby), or squamulose (scaly). They can also have different **reproductive structures**, such as soredia, isidia, apothecia, or pycnidia.
- Lichens have various **roles** in the ecosystem, such as **nitrogen fixation**, **soil formation**, **biomonitoring**, **bioindication**, **lichenometry**, and **food and medicine** for humans and animals.

Algae

- Algae are a large and diverse group of photosynthetic eukaryotic organisms that are typically aquatic and include seaweeds and many single-celled forms .
- Algae are classified into different groups based on their pigments, cell wall composition, flagella, storage products, and other features.
- Some of the major groups of algae are:
 - Green algae: They have chlorophyll a and b, cellulose in their cell walls, and store starch. They are closely related to plants and can be unicellular or multicellular. Examples: Spirogyra, Chlamydomonas, Ulva.
 - Brown algae: They have chlorophyll a and c, fucoxanthin (a brown pigment), alginic acid in their cell walls, and store laminarin. They are mostly marine and multicellular. Examples: Fucus, Sargassum, kelps.
 - Red algae: They have chlorophyll a and d, phycobilins (red pigments), agar and carrageenan in their cell walls, and store floridean starch. They are mostly marine and multicellular. Examples: Porphyra, Polysiphonia, coralline algae.

- Diatoms: They have chlorophyll a and c, fucoxanthin, silica in their cell walls, and store oil or chrysolaminarin. They are unicellular and have two overlapping shells. They are important producers in aquatic ecosystems. Examples: Cyclotella, Navicula, Pinnularia.
- Dinoflagellates: They have chlorophyll a and c, peridinin (a yellow pigment), cellulose plates in their cell walls, and store starch or oil. They are unicellular and have two flagella. Some of them are bioluminescent or cause red tides. Examples: Ceratium, Gonyaulax, Noctiluca.
- Euglenoids: They have chlorophyll a and b, paramylon in their cell walls, and store paramylon. They are unicellular and have one or two flagella. Some of them are heterotrophic or mixotrophic. Examples: Euglena, Phacus, Trachelomonas.
- Algae play an important role in the production of industrial chemicals and pharmaceuticals, such as:
 - Agar: A polysaccharide extracted from red algae that is used as a solidifying agent in microbiology, biotechnology, and food industry.
 - Carrageenan: A polysaccharide extracted from red algae that is used as a thickening, stabilizing, and gelling agent in food, cosmetics, and pharmaceuticals.
 - Alginic acid: A polysaccharide extracted from brown algae that is used as an emulsifier, stabilizer, and binder in food, textiles, and pharmaceuticals.
 - Iodine: A chemical element obtained from brown algae that is used as an antiseptic, disinfectant, and thyroid hormone supplement.
 - Beta-carotene: A pigment extracted from green algae that is used as a food colorant, antioxidant, and vitamin A precursor.
 - Spirulina: A blue-green alga that is used as a dietary supplement, protein source, and immune booster.

Protozoa

- Protozoa are **unicellular, microscopic, and primitive types of animals** that belong to the kingdom Protista .
- Protozoa are **heterotrophic**, meaning they use organic carbon as a source of energy. They also have a **true nucleus** and other membrane-bound organelles.
- Protozoa can be classified into four major groups based on their mode of locomotion :
 - **Amoeboid protozoa:** They move by pseudopodia (false feet) that extend and retract from the cell. Examples are Amoeba, Entamoeba, and Naegleria.
 - **Flagellated protozoa:** They move by one or more whip-like structures called flagella. Examples are Euglena, Trypanosoma, and Giardia.

- **Ciliated protozoa:** They move by numerous hair-like structures called cilia. Examples are Paramecium, Stentor, and Balantidium.
- **Sporozoan protozoa:** They do not have any locomotory organelles and are mostly parasitic. Examples are Plasmodium, Toxoplasma, and Cryptosporidium.
- Protozoa are found in various habitats, such as freshwater, marine, soil, and even inside other organisms as symbionts or parasites .
- Protozoa play important roles in the ecosystem, such as decomposing organic matter, recycling nutrients, and serving as food sources for other organisms .
- Protozoa can also cause diseases in humans and animals, such as malaria, sleeping sickness, dysentery, and leishmaniasis .

Helminthes

- Helminthes are **worm-like parasites** that can infect humans and other animals .
- They are **invertebrates** characterized by elongated, flat or round bodies .
- They are classified into three major groups based on their general external shape and the host organ they inhabit:
 - **Flukes (trematodes):** flat, leaf-shaped worms that live in the liver, lungs, blood vessels, or intestines of their hosts .
 - **Tapeworms (cestodes):** flat, ribbon-like worms that live in the intestines of their hosts and have a scolex (head) with hooks and suckers for attachment .
 - **Roundworms (nematodes):** cylindrical, thread-like worms that live in the intestines, blood, lymphatic system, or tissues of their hosts and have a complete digestive system .
- Helminthes have complex life cycles that involve one or more intermediate hosts and a definitive host .
- Helminthes can cause various diseases in humans, such as **schistosomiasis, taeniasis, ascariasis, filariasis, and trichinosis** .
- Helminthes can be diagnosed by microscopic examination of stool, urine, blood, or tissue samples, or by serological tests .
- Helminthes can be treated by anthelmintic drugs that target specific biochemical pathways or structures of the worms .
- Helminthes can be prevented by improving sanitation, hygiene, and vector control, and by avoiding raw or undercooked meat or fish that may contain larvae .

: **Helminths: Structure, Classification, Growth, and Development - Medical Microbiology - NC**
 What Are Helminths? - Verywell Health
 Parasitic worm - Wikipedia

Viral Structures

- Viruses are noncellular genetic elements that use a living cell for their replication and have an extracellular state.
- Viruses are ultramicroscopic particles containing nucleic acid (DNA or RNA) surrounded by a protein shell called a capsid .
- Some viruses have an outer envelope composed of lipids and proteins.
- Viruses are very diverse and can be classified by their shape, size, genome type, host range, and mode of replication.
- The basic structure of all viruses includes three main components:
 - Nucleic acid: The nucleic acid of a virus is found within its inner core that contains the genetic information for the synthesis of proteins and replication. The nucleic acid can be either DNA or RNA, single-stranded or double-stranded, linear or circular, segmented or non-segmented.
 - Capsid: The capsid is a protein layer or covering that forms a shell enclosing the genetic material of the virus. The capsid is composed of subunits called capsomeres, which can be identical or different. The capsid protects the nucleic acid from degradation and determines the shape of the virus.
 - Envelope: Many viruses have a lipoprotein bilayer that encloses the capsid. The envelope is derived from the host cell membrane and contains viral proteins called spikes or peplomers, which are involved in attachment and entry of the virus into the host cell . The envelope makes the virus more susceptible to environmental factors such as heat and detergents.
- The shape of the virus is determined by the arrangement of the capsomeres in the capsid. There are three main types of viral shapes:
 - Helical: The capsomeres are arranged in a helix around the nucleic acid, forming a rod-shaped or filamentous structure . Examples of helical viruses are tobacco mosaic virus, rabies virus, and influenza virus.
 - Icosahedral: The capsomeres are arranged in a symmetrical pattern of 20 triangular faces, forming a spherical structure . Examples of icosahedral viruses are poliovirus, adenovirus, and herpesvirus.
 - Complex: The capsomeres are arranged in a combination of helical and icosahedral structures, or have additional components such as tails, fibers, or envelopes . Examples of complex viruses are bacteriophages, poxviruses, and coronavirus.
- The size of the virus is measured by its diameter or length, and can range from 10 to 400 nanometers. The size of the virus is influenced by the type and amount of nucleic acid and the structure of the capsid and envelope. Generally, RNA viruses are smaller than DNA viruses, and enveloped viruses are larger than non-enveloped viruses.

Viral Multiplication

Viral multiplication is the process of producing new viruses using the host cell's machinery and resources. Viruses are infectious particles that consist of a nucleic acid genome (DNA or RNA) enclosed in a protein shell called a capsid. Some viruses also have a lipid envelope surrounding the capsid. Viruses can only replicate inside living cells, and they have a specific host range and tissue tropism, meaning they can only infect certain types of cells and organs.

The general steps of viral multiplication are:

- **Attachment:** The virus binds to specific receptors on the surface of the host cell, using proteins or glycoproteins on its capsid or envelope. This determines the specificity and susceptibility of the virus to the host cell.
- **Penetration:** The virus enters the host cell by different mechanisms, depending on the type of virus. Some viruses fuse their envelope with the host cell membrane and release their nucleocapsid into the cytoplasm. Some viruses are endocytosed by the host cell and escape from the endosome. Some viruses inject their genome into the host cell, leaving the capsid outside.
- **Uncoating:** The virus releases its genome from the capsid, making it available for transcription and replication. This may occur in the cytoplasm or in the nucleus, depending on the type of virus and its genome.
- **Biosynthesis:** The virus uses the host cell's enzymes, ribosomes, and other components to synthesize its own proteins and nucleic acids. The viral genome is transcribed into mRNA, which is translated into viral proteins. The viral genome is also replicated, using either the host cell's DNA or RNA polymerases, or the virus's own polymerases.
- **Assembly:** The newly synthesized viral components are assembled into new virions, either in the cytoplasm or in the nucleus, depending on the type of virus. The assembly process may involve the formation of a capsid around the genome, the acquisition of an envelope from the host cell membrane, and the incorporation of viral proteins into the capsid or envelope.
- **Release:** The new virions exit the host cell by different mechanisms, depending on the type of virus. Some viruses cause the host cell to lyse, breaking the cell membrane and releasing the virions. Some viruses bud from the host cell membrane, taking a portion of the membrane as their envelope. Some viruses are exocytosed by the host cell, using vesicles to transport the virions to the cell surface.

The viral multiplication cycle can be classified into two types, based on the fate of the host cell and the viral genome:

- **Lytic cycle:** The virus replicates and produces new virions that kill the host cell upon release. The viral genome does not integrate into the host cell's genome, and the infection is usually acute and self-limiting. Examples of viruses that undergo the lytic cycle are influenza virus, poliovirus, and rhinovirus.

- **Lysogenic cycle:** The virus integrates its genome into the host cell's genome, forming a provirus that remains latent and does not produce new virions. The provirus can be activated by certain stimuli, such as stress, UV light, or chemicals, and enter the lytic cycle. The infection is usually chronic and persistent, and can lead to oncogenesis or transformation of the host cell. Examples of viruses that undergo the lysogenic cycle are herpesvirus, HIV, and hepatitis B virus.

Role of microorganisms in the production of industrial chemicals and pharmaceuticals

- Microorganisms are used in industry to produce a variety of organic compounds including acids, growth stimulants, and enzymes.
- Microorganisms can be genetically modified or engineered to aid in large-scale production of desired products.
- Microorganisms can also use carbon dioxide and hydrogen as substrates to produce industrial chemicals and fuels.
- Some examples of industrial chemicals and pharmaceuticals produced by microorganisms are:
 - Amino acids: Microbes make only the L-isomer of amino acids, which are essential for protein synthesis and human nutrition. Some amino acids, such as lysine and glutamic acid, are used as food additives and flavor enhancers .
 - Citric acid: Produced by the mold *Aspergillus niger*, citric acid is widely used as a preservative, flavoring agent, and acidulant in food and beverage industries. It is also used in pharmaceuticals, cosmetics, and detergents .
 - Enzymes: Enzymes are biological catalysts that speed up chemical reactions. Microbes produce various enzymes, such as amylases, proteases, lipases, and cellulases, that are used in industries such as starch, textile, leather, paper, and biofuel .
 - Antibiotics: Antibiotics are substances that inhibit the growth or kill other microorganisms. They are used to treat bacterial infections and prevent diseases. Most antibiotics are produced by fungi or bacteria, such as penicillin by *Penicillium*, streptomycin by *Streptomyces*, and tetracycline by *Streptomyces aureofaciens* .
 - Vaccines: Vaccines are preparations that stimulate the immune system to produce antibodies against specific pathogens. They are used to prevent or reduce the severity of infectious diseases. Some vaccines are made from attenuated or killed microorganisms, such as polio vaccine from poliovirus, or from recombinant DNA technology, such as hepatitis B vaccine from yeast cells .

- Vitamins: Vitamins are organic compounds that are essential for normal metabolism and growth. Some vitamins, such as vitamin B12 and riboflavin, are produced by microorganisms and used as supplements or fortifiers in food and animal feed .

Unit 2 - Functional Anatomy of Prokaryotic and Eukaryotic Cells: Size, shape, and arrangement of bacterial cells. Structure and function of cells

- Prokaryotic and eukaryotic cells are the two main types of cells that make up all living organisms.
- Prokaryotic cells are simpler and smaller than eukaryotic cells. They lack a nucleus and other membrane-bound organelles. They have a single circular chromosome and plasmids that carry extra genetic information. They also have ribosomes, a cell wall, a cell membrane, and sometimes a capsule, flagella, pili, or fimbriae.
- Eukaryotic cells are more complex and larger than prokaryotic cells. They have a nucleus that contains linear chromosomes and a nucleolus that produces ribosomes. They also have various membrane-bound organelles, such as mitochondria, chloroplasts, endoplasmic reticulum, Golgi apparatus, lysosomes, peroxisomes, vacuoles, and vesicles. They have a cell membrane, a cytoskeleton, and sometimes a cell wall, cilia, or flagella.
- Bacterial cells are prokaryotic cells that can have different sizes, shapes, and arrangements. The size of bacterial cells ranges from 0.2 to 10 micrometers in diameter. The shape of bacterial cells can be classified into three main categories: cocci (spherical), bacilli (rod-shaped), and spirilla (spiral-shaped). The arrangement of bacterial cells can be determined by their pattern of division and attachment. Some common arrangements are: diplococci (pairs of cocci), streptococci (chains of cocci), staphylococci (clusters of cocci), diplobacilli (pairs of bacilli), streptobacilli (chains of bacilli), and spirochetes (flexible spirilla).
- The structure and function of cells are related to their adaptation to different environments and functions. For example, prokaryotic cells have a cell wall that protects them from osmotic pressure and provides shape and rigidity. Some prokaryotic cells have a capsule that helps them adhere to surfaces and evade the immune system. Some prokaryotic cells have flagella that enable them to move and sense their surroundings. Some prokaryotic cells have pili or fimbriae that facilitate attachment and genetic exchange. Eukaryotic cells have a nucleus that separates the genetic material from the cytoplasm and regulates gene expression. Eukaryotic cells have mitochondria that produce energy through cellular respiration. Eukaryotic cells have chloroplasts that perform photosynthesis and produce organic molecules. Eukaryotic cells have endoplasmic reticulum and Golgi apparatus that synthesize and modify proteins and lipids. Eukary-

otic cells have lysosomes and peroxisomes that degrade and detoxify waste and harmful substances. Eukaryotic cells have vacuoles and vesicles that store and transport materials. Eukaryotic cells have a cytoskeleton that provides support and movement. Eukaryotic cells have cilia or flagella that enable them to move and sense their surroundings.

Functional Anatomy of Prokaryotic and Eukaryotic Cells

- Prokaryotic and eukaryotic cells are the two basic types of cells that make up all living organisms.
- Prokaryotic cells are simpler and smaller than eukaryotic cells, and lack a nucleus and other membrane-bound organelles.
- Eukaryotic cells are more complex and larger than prokaryotic cells, and have a nucleus and other membrane-bound organelles that perform specific functions.
- The main differences between prokaryotic and eukaryotic cells are summarized in the table below:

Feature	Prokaryotic cells	Eukaryotic cells
Size	0.1-5 micrometers	10-100 micrometers
Shape	Mostly spherical, rod-shaped, or spiral	Various shapes, including spherical, rod-shaped, spiral, and irregular
Arrangement	Single-celled or in chains, clusters, or pairs	Single-celled or multicellular
Nucleus	Absent	Present
DNA	Circular, in the cytoplasm	Linear, in the nucleus
Histones	Absent	Present
Ribosomes	70S	80S
Cell division	Binary fission	Mitosis or meiosis
Chromosomes	One, usually	Two or more, usually
Organelles	None, except for ribosomes	Many, such as mitochondria, chloroplasts, endoplasmic reticulum, Golgi apparatus, lysosomes, etc.

Feature	Prokaryotic cells	Eukaryotic cells
Cell wall	Present, made of peptidoglycan (in bacteria) or pseudopeptidoglycan (in archaea)	Absent in animals, present in plants, fungi, and some protists, made of cellulose, chitin, or other polysaccharides
Plasma membrane	Present, with no sterols	Present, with sterols
Flagella	Present, simple and rigid	Present, complex and flexible
Cilia	Absent	Present, in some eukaryotes
Pili	Present, in some prokaryotes	Absent

- Prokaryotic and eukaryotic cells have some common features, such as a plasma membrane, cytoplasm, ribosomes, and genetic material (DNA or RNA).
- Prokaryotic and eukaryotic cells also have different functions and roles in living organisms. Prokaryotic cells are the main agents of biogeochemical cycles, decomposition, fermentation, nitrogen fixation, and pathogenesis. Eukaryotic cells are the main components of multicellular organisms, such as animals, plants, fungi, and protists, and are involved in various processes, such as photosynthesis, respiration, reproduction, differentiation, and communication.

Size, shape, and arrangement of bacterial cells

- Bacterial cells are microscopic organisms that cannot be seen by the naked eye. They are measured in micrometers (μm), which are one-thousandth of a millimeter. The average size of a bacterial cell varies between 0.2 and 2.0 μm .
- Bacterial cells have different shapes, such as spherical (cocci), rod-shaped (bacilli), spiral (spirilla), comma-shaped (vibrios), and corkscrew-shaped (spirochetes). Some bacteria can change their shape depending on the environment, such as pleomorphic bacteria.
- Bacterial cells can also arrange themselves in different patterns, such as single, pairs, chains, clusters, packets, and palisades. The arrangement depends on the plane of division, the number of divisions, and the degree of separation of the cells after division.
- Some examples of bacterial shapes and arrangements are:

- Cocci: spherical bacteria that can be single, diplococci (pairs), streptococci (chains), staphylococci (clusters), tetrads (packets of four), or sarcinae (packets of eight) .
 - Bacilli: rod-shaped bacteria that can be single, diplobacilli (pairs), streptobacilli (chains), or coccobacilli (short and oval) .
 - Spirilla: spiral bacteria that can be rigid or flexible, and can have one or more twists. They can be single or in pairs .
 - Vibrios: comma-shaped bacteria that can be single or in pairs .
 - Spirochetes: corkscrew-shaped bacteria that have a flexible cell wall and move by rotating. They can be single or in pairs .
- The size, shape, and arrangement of bacterial cells are important for their identification, classification, and function. For example, the shape and arrangement of bacteria can affect their surface area, nutrient uptake, motility, attachment, and pathogenicity .

Unit 2 - Functional Anatomy of Prokaryotic and Eukaryotic Cells: Size, shape, and arrangement of bacterial cells

- Bacterial cells are microscopic organisms that cannot be seen by the naked eye. They are measured in micrometers (μm), which are one-thousandth of a millimeter. The average size of a bacterial cell varies between 0.2 and 2.0 μm .
- Bacterial cells have different shapes and arrangements that help them adapt to different environments and perform different functions. The most common shapes are cocci (spherical), bacilli (rod-shaped), and spirilla (spiral-shaped) .
- Cocci are spherical bacteria that can be arranged in different patterns, such as:
 - Diplococci: pairs of cocci, e.g. *Neisseria gonorrhoeae*, the cause of gonorrhea.
 - Streptococci: chains of cocci, e.g. *Streptococcus pyogenes*, the cause of strep throat.
 - Staphylococci: clusters of cocci, e.g. *Staphylococcus aureus*, the cause of skin infections.
 - Tetrads: groups of four cocci, e.g. *Micrococcus luteus*, a soil bacterium.
 - Sarcinae: groups of eight cocci, e.g. *Sarcina ventriculi*, a stomach bacterium.
- Bacilli are rod-shaped bacteria that can be arranged in different patterns, such as:
 - Single bacilli: individual rods, e.g. *Escherichia coli*, a common gut bacterium.

- Diplobacilli: pairs of rods, e.g. *Klebsiella pneumoniae*, a cause of pneumonia.
- Streptobacilli: chains of rods, e.g. *Bacillus anthracis*, the cause of anthrax.
- Coccobacilli: short and oval rods, e.g. *Haemophilus influenzae*, a cause of meningitis.
- Palisades: parallel or V-shaped arrangements of rods, e.g. *Corynebacterium diphtheriae*, the cause of diphtheria.
- Spirilla are spiral-shaped bacteria that can be arranged in different patterns, such as:
 - Vibrios: curved or comma-shaped rods, e.g. *Vibrio cholerae*, the cause of cholera.
 - Spirilla: rigid and helical rods, e.g. *Spirillum minus*, the cause of rat-bite fever.
 - Spirochetes: flexible and corkscrew-shaped rods, e.g. *Treponema pallidum*, the cause of syphilis.
- Some bacteria have other shapes and arrangements, such as:
 - Star-shaped bacteria, e.g. *Stella humosa*, a marine bacterium.
 - Square-shaped bacteria, e.g. *Haloquadratum walsbyi*, a salt-loving bacterium.
 - Filamentous bacteria, e.g. *Streptomyces coelicolor*, a soil bacterium that produces antibiotics.
 - Pleomorphic bacteria, e.g. *Mycoplasma pneumoniae*, a bacterium that lacks a cell wall and can change its shape.

Arrangement of Bacterial Cells

- Bacteria are microscopic, single-celled organisms that can have different shapes and arrangements.
- The shape of a bacterial cell is determined by its cell wall, which is a rigid layer of peptidoglycan that surrounds the cell membrane.
- The arrangement of a bacterial cell is determined by how it divides and how the daughter cells remain attached to each other.
- The most common shapes of bacterial cells are:
 - Cocci: spherical or oval cells
 - Bacilli: rod-shaped or cylindrical cells
 - Spirilla: spiral or curved cells
- The most common arrangements of bacterial cells are:
 - Diplo: pairs of cells
 - Staphylo: grape-like clusters of cells
 - Strepto: chains of cells
 - Tetrad: groups of four cells in a square
 - Sarcina: groups of eight cells in a cube
- These arrangements can be applied to the different shapes of bacterial cells, for example:

- Diplococci: pairs of cocci
- Streptobacilli: chains of bacilli
- Vibrio: comma-shaped spirilla
- The shape and arrangement of bacterial cells can help identify and classify them.

Structure and function of cells

- Cells are the basic units of life and the smallest living things that can perform all the functions of life.
- Cells can be classified into two major types: prokaryotic and eukaryotic.
- Prokaryotic cells are simpler and smaller than eukaryotic cells. They do not have a nucleus or other membrane-bound organelles. They usually have a single circular chromosome, a cell wall, a plasma membrane, ribosomes, and sometimes a flagellum or other appendages.
- Eukaryotic cells are more complex and larger than prokaryotic cells. They have a nucleus that contains the genetic material, and various membrane-bound organelles that perform different functions. They usually have multiple linear chromosomes, a cytoskeleton, a plasma membrane, ribosomes, and sometimes a cell wall or other structures.
- Some of the common organelles in eukaryotic cells are:
 - Mitochondria: the sites of cellular respiration, where energy is produced from glucose and oxygen.
 - Chloroplasts: the sites of photosynthesis, where light energy is converted into chemical energy in plants and some algae.
 - Endoplasmic reticulum: a network of membranes that transports and modifies proteins and lipids.
 - Golgi apparatus: a stack of flattened membranes that sorts and packages proteins and lipids for export or use within the cell.
 - Lysosomes: sacs of enzymes that digest and recycle materials within the cell.
 - Vacuoles: large sacs that store water, nutrients, waste, or other substances.
 - Cytoskeleton: a network of protein fibers that provides shape, support, and movement to the cell and its organelles.
 - Cilia and flagella: hair-like or whip-like structures that enable the cell or some of its organelles to move or sense the environment.
- The size, shape, and arrangement of bacterial cells can vary widely. Some common shapes are:
 - Coccus: spherical or oval
 - Bacillus: rod-shaped or cylindrical
 - Spirillum: spiral or curved
 - Vibrio: comma-shaped
 - Spirochete: flexible and helical
- Bacterial cells can be arranged in different patterns, such as:

- Single: individual cells
- Diplo: pairs of cells
- Strepto: chains of cells
- Staphylo: clusters of cells
- Tetrad: groups of four cells
- Sarcina: groups of eight cells
- Palisade: parallel rows of cells

Unit 3 - Catabolic & anabolic reactions: enzymes, energy production and carbohydrate metabolism. Lipid & protein catabolism, Energy production mechanism, metabolic diversity & pathways of energy use. Integration of metabolism

- Catabolic and anabolic reactions are two types of metabolic reactions that occur in living cells.
- Catabolic reactions break down complex molecules into simpler ones, releasing energy and reducing power .
- Anabolic reactions use energy and reducing power to build complex molecules from simpler ones .
- Catabolic and anabolic reactions are often coupled, with the energy and reducing power released from catabolism being used to drive anabolism .
- Enzymes are biological catalysts that speed up the rate of metabolic reactions by lowering the activation energy .
- Enzymes are highly specific for their substrates and can be regulated by various factors, such as temperature, pH, inhibitors, activators, and feedback mechanisms .
- Energy production is the process of converting the chemical energy stored in fuel molecules (such as glucose, fatty acids, and amino acids) into a more usable form, such as adenosine triphosphate (ATP) .
- Energy production involves three main stages: glycolysis, the citric acid cycle, and oxidative phosphorylation .
- Glycolysis is the breakdown of glucose into two molecules of pyruvate, generating two molecules of ATP and two molecules of NADH (a reducing agent) per glucose molecule .
- The citric acid cycle is a series of reactions that oxidize acetyl-CoA (derived from pyruvate or fatty acids) into carbon dioxide, generating two molecules of ATP, six molecules of NADH, and two molecules of FADH₂ (another reducing agent) per acetyl-CoA molecule .
- Oxidative phosphorylation is the process of transferring the electrons from NADH and FADH₂ to oxygen, forming water and generating a proton gradient across the inner mitochondrial membrane. This proton gradient drives the synthesis of ATP by the enzyme ATP synthase .

- Carbohydrate metabolism is the set of metabolic pathways that involve the synthesis and degradation of carbohydrates, such as sugars, starch, glycogen, and cellulose.
- Carbohydrate metabolism includes glycolysis, gluconeogenesis, glycogenesis, glycogenolysis, pentose phosphate pathway, and glyoxylate cycle.
- Gluconeogenesis is the synthesis of glucose from non-carbohydrate precursors, such as lactate, pyruvate, glycerol, and amino acids.
- Glycogenesis is the synthesis of glycogen from glucose, which is stored in the liver and muscles as a reserve of energy.
- Glycogenolysis is the breakdown of glycogen into glucose, which is released into the bloodstream to maintain blood glucose levels.
- Pentose phosphate pathway is a branch of carbohydrate metabolism that generates pentose sugars (such as ribose) for nucleic acid synthesis and NADPH for biosynthesis and detoxification.
- Glyoxylate cycle is a variation of the citric acid cycle that occurs in some plants and bacteria, which allows them to convert acetyl-CoA into carbohydrates.
- Lipid catabolism is the breakdown of lipids (such as triglycerides, phospholipids, and cholesterol) into smaller molecules, such as glycerol, fatty acids, and cholesterol derivatives .
- Lipid catabolism includes lipolysis, beta-oxidation, ketogenesis, and cholesterol metabolism .
- Lipolysis is the hydrolysis of triglycerides into glycerol and fatty acids, which can be used for energy production or biosynthesis .
- Beta-oxidation is the sequential removal of two-carbon units from the end of a fatty acid chain, forming acetyl-CoA, NADH, and FADH₂, which can

Catabolic and Anabolic Reactions

- Catabolic and anabolic reactions are two types of metabolic reactions that occur in living cells.
- Catabolic reactions are those that break down complex molecules into simpler ones, releasing energy in the process. Examples of catabolic reactions are glycolysis, cellular respiration, and lipid and protein catabolism.
- Anabolic reactions are those that build up complex molecules from simpler ones, using energy in the process. Examples of anabolic reactions are protein synthesis, photosynthesis, and lipid and carbohydrate synthesis.
- Catabolic and anabolic reactions are often coupled, meaning that the energy released by catabolic reactions is used to power anabolic reactions. A common molecule that mediates this coupling is adenosine triphosphate (ATP), which is produced by catabolic reactions and consumed by anabolic reactions.
- Catabolic and anabolic reactions are regulated by enzymes, which are proteins that catalyze specific chemical reactions. Enzymes can be activated or inhibited by various factors, such as temperature, pH, substrate

concentration, and feedback mechanisms.

- Catabolic and anabolic reactions are essential for maintaining the homeostasis and growth of living organisms. Catabolic reactions provide the energy and building blocks for anabolic reactions, while anabolic reactions synthesize the macromolecules and structures that make up the cells and tissues.

Enzymes

- Enzymes are **biological molecules** (typically proteins) that significantly speed up the rate of virtually all of the chemical reactions that take place within cells .
- Enzymes are **vital for life** and serve a wide range of important functions in the body, such as aiding in digestion and metabolism .
- Enzymes act as **catalysts**, which means they lower the activation energy needed for a reaction to occur, and they increase the rate of the reaction .
- Enzymes are **specific**, which means they only catalyze one type of reaction or substrate .
- Enzymes have an **active site**, which is the region where the substrate binds to the enzyme and the reaction takes place .
- Enzymes can be **regulated** by various factors, such as temperature, pH, inhibitors, activators, and feedback mechanisms .
- Enzymes are **classified** into six major groups based on the type of reaction they catalyze: oxidoreductases, transferases, hydrolases, lyases, isomerases, and ligases.

Energy production for the notes of the Unit 3 - Catabolic & anabolic reactions: enzymes, energy production and carbohydrate metabolism. Lipid & protein catabolism, Energy production mech- anism, metabolic diversity & pathways of energy use. Integration of metabolism in the subject of REMEDIAL BIOLOGY-II KCS

- Energy production is the process of converting different sources of energy, such as chemical, light, or nuclear energy, into a usable form, such as ATP, the universal energy currency of cells.
- Energy production involves various metabolic pathways, which are sequences of enzyme-catalyzed reactions that transform one or more starting molecules into products.
- Metabolic pathways can be classified into two types: catabolic and an-

abolic.

- Catabolic pathways are those that break down large molecules into smaller ones, releasing energy in the process. Examples of catabolic pathways are glycolysis, the citric acid cycle, and oxidative phosphorylation, which are involved in carbohydrate metabolism; beta-oxidation, which is involved in lipid metabolism; and proteolysis, which is involved in protein metabolism.
- Anabolic pathways are those that synthesize large molecules from smaller ones, requiring energy in the process. Examples of anabolic pathways are gluconeogenesis, the pentose phosphate pathway, and glycogenesis, which are involved in carbohydrate metabolism; fatty acid synthesis, which is involved in lipid metabolism; and protein synthesis, which is involved in protein metabolism.
- Enzymes are biological catalysts that speed up the rate of metabolic reactions by lowering the activation energy required for the reaction to occur. Enzymes are highly specific for their substrates and products, and can be regulated by various factors, such as temperature, pH, inhibitors, activators, and feedback mechanisms.
- Energy production mechanism is the process of coupling the energy released from catabolic pathways to the energy required for anabolic pathways, mainly through the formation and hydrolysis of ATP. ATP is composed of an adenine base, a ribose sugar, and three phosphate groups. The bonds between the phosphate groups are high-energy bonds that can be broken to release energy for cellular work. ATP can be synthesized by three main mechanisms: substrate-level phosphorylation, oxidative phosphorylation, and photophosphorylation.
 - Substrate-level phosphorylation is the direct transfer of a phosphate group from a substrate molecule to ADP, forming ATP. This occurs in some steps of glycolysis and the citric acid cycle, as well as in some anabolic pathways, such as gluconeogenesis and glycogenesis.
 - Oxidative phosphorylation is the indirect synthesis of ATP from ADP and inorganic phosphate, using the energy derived from the electron transport chain and the proton gradient across the inner mitochondrial membrane. This occurs in the final stage of carbohydrate, lipid, and protein metabolism, and is the major source of ATP production in aerobic organisms.
 - Photophosphorylation is the synthesis of ATP from ADP and inorganic phosphate, using the energy derived from the light-dependent reactions of photosynthesis. This occurs in the thylakoid membranes of chloroplasts, and is the source of ATP production in photosynthetic organisms.
- Metabolic diversity is the variation in the types and pathways of metabolism among different organisms, depending on their environmental conditions, nutritional requirements, and evolutionary history. Some examples of metabolic diversity are:
 - Fermentation is the partial breakdown of organic molecules without

oxygen, producing various end products, such as ethanol, lactate, or acetate, and a small amount of ATP. Fermentation occurs in some bacteria, fungi, and animal cells, such as muscle cells, when oxygen is scarce or absent.

- Anaerobic respiration is the complete breakdown of organic molecules without oxygen, using an alternative electron acceptor, such as nitrate, sulfate, or carbon dioxide, and producing a moderate amount of ATP. Anaerobic respiration occurs in some bacteria and archaea, such as methanogens, sulfate-reducers, and denitrifiers, that live in oxygen-poor or oxygen-free environments.
- Chemosynthesis is the synthesis of organic molecules from inorganic molecules, using the energy derived from the oxidation of reduced compounds, such as hydrogen sulfide, ammonia, or iron. Chemosynthesis occurs in some bacteria and archaea, such as sulfur-oxidizers, nitrifiers, and iron-oxidizers, that live in extreme environments, such as hydrothermal vents, volcanic soils, or acidic springs.
- Photosynthesis is the synthesis of organic molecules from carbon dioxide and water, using the energy derived from light. Photosynthesis occurs in plants, algae

Carbohydrate Metabolism

- Carbohydrate metabolism is the whole of the biochemical processes responsible for the metabolic formation, breakdown, and interconversion of carbohydrates in living organisms.
- Carbohydrates are organic molecules composed of carbon, hydrogen, and oxygen atoms. They can be classified into monosaccharides (simple sugars), disaccharides (two monosaccharides linked by a glycosidic bond), oligosaccharides (three to ten monosaccharides), and polysaccharides (more than ten monosaccharides).
- The most common monosaccharides are glucose, fructose, and galactose. The most common disaccharides are sucrose (glucose + fructose), lactose (glucose + galactose), and maltose (glucose + glucose). The most common polysaccharides are starch, glycogen, and cellulose.
- Carbohydrate metabolism begins with digestion in the small intestine where monosaccharides are absorbed into the blood stream. Blood sugar concentrations are controlled by three hormones: insulin, glucagon, and epinephrine. Insulin lowers blood glucose levels by stimulating the uptake of glucose by cells and the storage of glucose as glycogen in the liver and muscles. Glucagon raises blood glucose levels by stimulating the breakdown of glycogen and the release of glucose by the liver. Epinephrine also raises blood glucose levels by stimulating the breakdown of glycogen and the release of glucose by the liver and muscles, as well as the breakdown of fat and protein for energy.
- The main pathway of carbohydrate metabolism is glycolysis, which is the

breakdown of glucose into two molecules of pyruvate. Glycolysis occurs in the cytoplasm of all cells and does not require oxygen. It produces a net gain of two molecules of ATP (adenosine triphosphate) and two molecules of NADH (nicotinamide adenine dinucleotide) per glucose molecule. Pyruvate can then enter different pathways depending on the availability of oxygen and the type of cell.

- If oxygen is present, pyruvate can enter the mitochondria and be converted into acetyl-CoA (acetyl coenzyme A) by the pyruvate dehydrogenase complex. Acetyl-CoA then enters the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle), which is a series of reactions that produce two molecules of CO₂ (carbon dioxide), one molecule of ATP, three molecules of NADH, and one molecule of FADH₂ (flavin adenine dinucleotide) per acetyl-CoA molecule. The NADH and FADH₂ then enter the electron transport chain, which is a series of protein complexes that transfer electrons from NADH and FADH₂ to O₂ (oxygen), producing H₂O (water) and a large amount of ATP. This process is called cellular respiration or aerobic respiration.
- If oxygen is absent or limited, pyruvate can undergo anaerobic respiration or fermentation. In anaerobic respiration, pyruvate is converted into lactate (lactic acid) by lactate dehydrogenase, which regenerates NAD⁺ (nicotinamide adenine dinucleotide) from NADH. This allows glycolysis to continue, but produces less ATP and more lactate, which can cause muscle fatigue and soreness. In fermentation, pyruvate is converted into ethanol (alcohol) and CO₂ by pyruvate decarboxylase and alcohol dehydrogenase, which also regenerate NAD⁺ from NADH. This process occurs in some microorganisms, such as yeast, and is used for the production of alcoholic beverages and bread.
- Carbohydrate metabolism can also involve the interconversion of different carbohydrates, such as the conversion of glucose into galactose or fructose, or the conversion of glucose into ribose or deoxyribose. These processes are catalyzed by specific enzymes, such as hexokinase, phosphoglucose isomerase, phosphofructokinase, aldolase, transketolase, and transaldolase.
- Carbohydrate metabolism can also be integrated with lipid and protein metabolism, as some intermediates of glycolysis and the citric acid cycle can be used for the

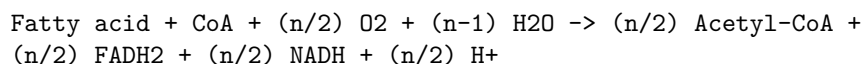
Lipid and Protein Catabolism

- Lipid and protein catabolism are the processes of breaking down lipids and proteins, respectively, into smaller molecules that can be used for energy production or biosynthesis.
- Lipids and proteins are important macromolecules that store energy, provide structural support, and perform various functions in living organisms.
- Lipid and protein catabolism involve different enzymes, pathways, and intermediates than carbohydrate catabolism, but they ultimately converge

with the central catabolic pathways of glycolysis, the citric acid cycle, and oxidative phosphorylation.

Lipid Catabolism

- Lipids are a diverse group of organic molecules that include fats, oils, phospholipids, steroids, and waxes.
- The most common form of lipid catabolism is the breakdown of triglycerides, which are composed of glycerol and three fatty acids.
- Triglycerides are hydrolyzed by lipases, enzymes that cleave the ester bonds between glycerol and fatty acids .
- Glycerol can be converted to dihydroxyacetone phosphate (DHAP), an intermediate of glycolysis, by a series of reactions involving glycerol kinase and glycerol-3-phosphate dehydrogenase .
- Fatty acids are activated by coenzyme A (CoA) and transported into the mitochondrial matrix by the carnitine shuttle, a system that exchanges fatty acyl-CoA for carnitine across the inner mitochondrial membrane .
- Fatty acyl-CoA undergoes beta-oxidation, a cyclic process that removes two-carbon units from the carboxyl end of the fatty acid chain and produces acetyl-CoA, NADH, and FADH₂ .
- Acetyl-CoA enters the citric acid cycle, where it is oxidized to carbon dioxide and water, generating more NADH and FADH₂ .
- NADH and FADH₂ donate their electrons to the electron transport chain, where they drive the synthesis of ATP by oxidative phosphorylation .
- The overall equation for the complete oxidation of a fatty acid is:



where n is the number of carbon atoms in the fatty acid chain.

Protein Catabolism

- Proteins are polymers of amino acids, which are linked by peptide bonds.
- Protein catabolism begins with the hydrolysis of proteins into amino acids by proteases, enzymes that cleave the peptide bonds .
- Amino acids are transported into the cells by specific transporters and then deaminated, a process that removes the amino group from the amino acid and produces ammonia and an organic acid .
- Ammonia is toxic to cells and must be eliminated or converted to a less harmful form, such as urea or uric acid, depending on the organism .
- The organic acids derived from amino acids can enter different points of the central catabolic pathways, depending on their structure .
- Some organic acids can be converted to pyruvate, the end product of glycolysis, by transamination or decarboxylation reactions .
- Some organic acids can be converted to acetyl-CoA, the entry point of the citric acid cycle, by oxidative decarboxylation or thiolysis reactions .

- Some organic acids can be converted to intermediates of the citric acid cycle, such as oxaloacetate, alpha-ketoglutarate, succinyl-CoA, fumarate, or malate, by various reactions involving transamination, dehydrogenation, or hydration .
- The overall equation for the complete oxidation of an amino

Energy production mechanism

- Energy production is the process of converting energy from one form to another in living systems.
- Energy comes to living systems through electrons occupying high energy states, either from food (respiratory chains) or from light (photosynthesis).
- Energy is exchanged between living systems and their surroundings as they use energy from the sun to perform photosynthesis or consume energy-storing molecules and release energy to the environment by doing work and releasing heat.
- Energy is subject to physical laws, such as the first law of thermodynamics, which states that energy cannot be created or destroyed, only transformed or transferred, and the second law of thermodynamics, which states that the entropy (disorder) of the universe tends to increase.
- Energy is stored in the chemical bonds of organic molecules, such as carbohydrates, lipids, and amino acids, which are used as fuel and building material by living systems.
- Energy can be released from organic molecules by breaking down their bonds through a series of chemical reactions called metabolic pathways, which can be either catabolic (breaking down molecules and releasing energy) or anabolic (building up molecules and consuming energy).
- The primary energy-releasing pathways of metabolism involve the breakdown and synthesis of carbohydrates, lipids, and amino acids. The most common pathway for breaking down carbohydrates is glycolysis, which converts glucose into pyruvate and produces a net gain of two molecules of ATP (adenosine triphosphate), the main energy currency of the cell .
- The pyruvate produced by glycolysis can be further oxidized in the presence of oxygen by the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle), which generates more ATP and high-energy electron carriers, such as NADH (nicotinamide adenine dinucleotide) and FADH₂ (flavin adenine dinucleotide).
- The high-energy electron carriers produced by the citric acid cycle can donate their electrons to the electron transport chain, a series of protein complexes embedded in the inner membrane of the mitochondria (the organelles responsible for cellular respiration). The electron transport chain uses the energy from the electrons to pump protons across the membrane, creating a proton gradient that drives the synthesis of ATP by the enzyme ATP synthase.
- The final electron acceptor in the electron transport chain is oxygen, which

combines with protons to form water. This process is called oxidative phosphorylation, and it produces the majority of ATP in aerobic (oxygen-requiring) organisms.

- In the absence of oxygen, pyruvate can be reduced to lactate or ethanol by the process of fermentation, which regenerates NAD⁺ (nicotinamide adenine dinucleotide) from NADH, allowing glycolysis to continue. Fermentation produces much less ATP than oxidative phosphorylation, and it results in the accumulation of waste products, such as lactic acid or alcohol.
- Lipids and amino acids can also be broken down for energy by different metabolic pathways. Lipids can be hydrolyzed into glycerol and fatty acids, which can enter glycolysis and the citric acid cycle, respectively. Amino acids can be deaminated (removal of the amino group) and converted into intermediates of glycolysis or the citric acid cycle, depending on their chemical structure.
- The synthesis of organic molecules from simpler precursors is called biosynthesis or anabolism, and it requires energy input from ATP or other high-energy molecules. Biosynthesis allows living systems to grow, repair, and maintain their structures, as well as to store energy for future use.
- The integration of metabolism is the coordination and regulation of the various metabolic pathways to meet the changing needs and conditions of the organism. The integration of metabolism is achieved by various mechanisms, such as feedback inhibition, allosteric regulation, hormonal control, and gene expression.
- The metabolic diversity of living systems reflects the adaptation to different environmental and ecological niches. Different organisms can use different sources of energy and carbon, such as light, organic molecules, or inorganic molecules, and can produce different types of waste products, such as oxygen, carbon dioxide, methane, or sulfur. The metabolic diversity of living systems is the basis of the biogeochemical cycles that shape the Earth's biosphere.

Metabolic Diversity and Pathways of Energy Use

- Metabolism is the sum of all the chemical reactions that occur in a living organism to maintain its life functions, such as growth, reproduction, and response to stimuli.
- Metabolic diversity mainly refers to the different metabolic strategies that organisms have evolved to obtain energy from various sources, such as light, organic molecules, or inorganic molecules.
- Energy is the capacity to do work, and it can be stored in different forms, such as chemical, mechanical, electrical, or thermal energy.
- Energy metabolism is the general process by which living cells acquire and use the energy needed to stay alive, to grow, and to reproduce.
- Energy metabolism involves two types of reactions: catabolic and anabolic

reactions.

- Catabolic reactions are those that break down complex molecules into simpler ones, releasing energy in the process. For example, the breakdown of glucose into carbon dioxide and water is a catabolic reaction that releases chemical energy that can be used to make ATP, the universal energy currency of cells.
- Anabolic reactions are those that build up complex molecules from simpler ones, consuming energy in the process. For example, the synthesis of proteins from amino acids is an anabolic reaction that requires energy from ATP.
- The energy released or consumed in a chemical reaction can be measured by its change in free energy, which is the portion of energy that is available to do work. The change in free energy is denoted by ΔG , and it can be calculated by the following equation: $\Delta G = \Delta H - T\Delta S$, where ΔH is the change in enthalpy (heat content), T is the absolute temperature, and ΔS is the change in entropy (disorder).
 - A reaction is spontaneous (or exergonic) if it has a negative ΔG , meaning that it releases free energy. A reaction is non-spontaneous (or endergonic) if it has a positive ΔG , meaning that it requires free energy. A reaction is at equilibrium if it has a zero ΔG , meaning that it does not change the free energy of the system.
- Cells use different metabolic pathways, or types of chemical reactions, to produce the right kind of energy to fuel different activities. The metabolic pathways can be broadly divided into three categories based on how quickly they need energy and how much of it they need:
 - The phosphagen system (or ATP-PC system) is the fastest and simplest way to produce energy, but it can only provide a small amount of energy for a short duration (about 10 seconds). It involves the use of stored ATP and creatine phosphate (CP) in the muscle cells to regenerate ATP from ADP. This system is used for high-intensity and short-duration activities, such as sprinting or lifting weights.
 - Glycolysis is the next fastest way to produce energy, but it can only provide a moderate amount of energy for a moderate duration (about 2 minutes). It involves the breakdown of glucose (from blood or glycogen) into pyruvate, producing 2 ATP and 2 NADH per glucose molecule. This system can operate in the presence or absence of oxygen. In the presence of oxygen, pyruvate can enter the mitochondria and be further oxidized in the citric acid cycle and the electron transport chain, producing more ATP and water. This is called aerobic glycolysis. In the absence of oxygen, pyruvate is converted into lactate, which can be recycled back to glucose in the liver or used by other tissues. This is called anaerobic glycolysis. This system is used for moderate-intensity and moderate-duration activities, such as jogging or cycling.
 - The aerobic system is the slowest but most efficient way to produce energy, and it can provide a large amount of energy for a long du-

ration (more than 2 minutes). It involves the complete oxidation of glucose, fatty acids, or amino acids in the mitochondria, producing 36 to 38 ATP per glucose molecule, 129 ATP per fatty acid molecule, and variable ATP per amino acid molecule. This system requires oxygen and produces carbon dioxide and water as byproducts. This system is used for low-intensity and long-duration activities, such as walking or resting.

Integration of metabolism

- Metabolism is the sum of all the chemical reactions that occur in a living cell to maintain its structure and function.
- Metabolism can be divided into two major types: catabolism and anabolism.
 - Catabolism is the breakdown of complex molecules into simpler ones, releasing energy and reducing power.
 - Anabolism is the synthesis of complex molecules from simpler ones, requiring energy and reducing power.
- Metabolism is regulated by various factors, such as enzymes, hormones, substrates, cofactors, and feedback mechanisms.
- Metabolism is integrated in different tissues and organs to coordinate the energy and biosynthetic needs of the whole organism.
 - Three hormones—insulin, glucagon, and epinephrine—play a critical role in integrating metabolism, especially energy metabolism, in different tissues.
 - Insulin is secreted by the pancreas in response to high blood glucose levels and stimulates glucose uptake, glycolysis, glycogen synthesis, and fatty acid synthesis in the liver, muscle, and adipose tissue.
 - Glucagon is secreted by the pancreas in response to low blood glucose levels and stimulates glycogenolysis, gluconeogenesis, and ketogenesis in the liver, and lipolysis in the adipose tissue.
 - Epinephrine is secreted by the adrenal glands in response to stress and stimulates glycogenolysis, gluconeogenesis, and lipolysis in the liver, muscle, and adipose tissue, and increases the heart rate and blood pressure.
- Metabolism is also integrated in different cellular compartments, such as the cytosol, the mitochondria, the endoplasmic reticulum, and the peroxisomes, to facilitate the transport and exchange of metabolites and cofactors.
 - The cytosol is the site of glycolysis, pentose phosphate pathway, fatty acid synthesis, and nucleotide synthesis.
 - The mitochondria is the site of the citric acid cycle, oxidative phosphorylation, fatty acid oxidation, and amino acid catabolism.
 - The endoplasmic reticulum is the site of lipid synthesis, detoxification, and calcium storage.

- The peroxisomes are the site of fatty acid oxidation, hydrogen peroxide degradation, and bile acid synthesis.
- Metabolism is also integrated in different metabolic pathways, such as glycolysis, gluconeogenesis, glycogen synthesis, glycogen degradation, pentose phosphate pathway, citric acid cycle, oxidative phosphorylation, fatty acid synthesis, fatty acid oxidation, ketogenesis, cholesterol synthesis, amino acid synthesis, amino acid degradation, urea cycle, and nucleotide synthesis.
 - These pathways are interconnected by common intermediates, such as glucose, pyruvate, acetyl-CoA, oxaloacetate, and citrate.
 - These pathways are also regulated by common mechanisms, such as allosteric control, covalent modification, transcriptional control, and hormonal control.
 - These pathways are also coordinated by common goals, such as maintaining the energy balance, the redox balance, and the biosynthetic balance of the cell.

Unit 4 - Energy Utilization

Structure of mitochondria

- Mitochondria are small, rod-shaped organelles found in both plant and animal cells .
- They have two membranes, an outer membrane and an inner membrane, made of phospholipid layers .
- The outer membrane is smooth and encloses the organelle, while the inner membrane is folded into projections called cristae .
- The cristae increase the surface area of the inner membrane, where the reactions of cellular respiration take place.
- The space between the two membranes is called the intermembrane space, and the space inside the inner membrane is called the matrix .
- The matrix contains enzymes, ribosomes, DNA, and various molecules involved in energy production .
- Mitochondria are the sites of aerobic respiration, which generates adenosine triphosphate (ATP), the main source of chemical energy for the cell .
- Mitochondria have their own DNA and ribosomes, and can replicate independently of the cell .
- Mitochondria are thought to have evolved from ancient bacteria that were engulfed by ancestral eukaryotic cells, forming a symbiotic relationship . This is known as the endosymbiotic theory.

Cellular respiration

- Cellular respiration is the process of breaking down organic molecules, such as glucose, to release energy in the form of ATP .
- Cellular respiration can be divided into four stages: glycolysis, formation of acetyl coenzyme A (acetyl-CoA), Krebs cycle, and electron transport chain (ETC) and oxidative phosphorylation .
- Glycolysis is the first stage of cellular respiration, and it occurs in the cytoplasm of the cell .
- Glycolysis involves the splitting of one molecule of glucose (a six-carbon sugar) into two molecules of pyruvate (a three-carbon compound), with the net production of two ATP and two NADH (a reduced form of nicotinamide adenine dinucleotide, an electron carrier) .
- Pyruvate then enters the mitochondria, where it is converted into acetyl-CoA, a two-carbon compound that is attached to coenzyme A, a carrier molecule .
- This conversion also produces one NADH and one carbon dioxide (CO_2) per pyruvate molecule .
- Acetyl-CoA then enters the Krebs cycle, also known as the citric acid cycle or the tricarboxylic acid (TCA) cycle, which occurs in the matrix of the mitochondria .
- The Krebs cycle is a series of reactions that involves the oxidation of acetyl-CoA and the generation of various intermediates, such as citrate, succinate, and oxaloacetate .
- The Krebs cycle also produces two CO_2 , one ATP, three NADH, and one FADH₂ (a reduced form of flavin adenine dinucleotide, another electron carrier) per acetyl-CoA molecule .
- The NADH and FADH₂ produced in glycolysis, pyruvate oxidation, and the Krebs cycle then donate their electrons to the ETC, which is located in the inner membrane of the mitochondria .
- The ETC is a series of protein complexes and electron carriers that transfer the electrons from NADH and FADH₂ to the final electron acceptor, which is oxygen (O_2) .
- The ETC also pumps protons (H^+) from the matrix to the intermembrane space, creating a proton gradient across the inner membrane .
- The proton gradient drives the synthesis of ATP by a protein complex called ATP synthase, which is embedded in the inner membrane .
- This process of coupling the ETC and ATP synthesis is called oxidative phosphorylation, and it produces

Energy Utilization

- Energy utilization is the process of converting the chemical energy stored in organic molecules into other forms of energy that can be used by cells for various functions.

- Energy utilization involves a series of biochemical reactions that take place in different cellular compartments, such as the cytoplasm and the mitochondria.
- Energy utilization is mainly based on the oxidation of carbohydrates, such as glucose, which can be derived from the digestion of food or the breakdown of glycogen.
- The oxidation of glucose involves three major stages: glycolysis, the citric acid cycle (or Krebs cycle), and the electron transport chain (or oxidative phosphorylation) .
- In glycolysis, glucose is split into two molecules of pyruvate, generating a net gain of two molecules of ATP (adenosine triphosphate) and two molecules of NADH (nicotinamide adenine dinucleotide, reduced form) .
- In the presence of oxygen, pyruvate is transported into the mitochondria, where it is converted into acetyl-CoA (acetyl coenzyme A) by a complex of enzymes called the pyruvate dehydrogenase complex . This reaction also produces one molecule of CO₂ (carbon dioxide) and one molecule of NADH per pyruvate .
- Acetyl-CoA then enters the citric acid cycle, where it is oxidized to CO₂, generating one molecule of ATP, three molecules of NADH, and one molecule of FADH₂ (flavin adenine dinucleotide, reduced form) per cycle .
- The NADH and FADH₂ produced in glycolysis and the citric acid cycle are then transferred to the electron transport chain, which is located in the inner membrane of the mitochondria .
- The electron transport chain consists of a series of protein complexes and electron carriers that transfer the electrons from NADH and FADH₂ to the final acceptor, which is oxygen . This process releases energy that is used to pump protons (H⁺) across the inner membrane, creating a proton gradient .
- The proton gradient drives the synthesis of ATP by a protein complex called ATP synthase, which is also located in the inner membrane of the mitochondria . This process is called oxidative phosphorylation, because it involves the coupling of oxygen reduction and ATP production .
- ATP is the main energy currency of the cell, as it can be used to power various cellular processes, such as muscle contraction, nerve transmission, biosynthesis, and transport .
- The oxidation of glucose is not the only source of energy for the cell, as other organic molecules, such as fats and proteins, can also be metabolized to generate ATP. However, the oxidation of glucose is the most efficient and versatile pathway, as it can operate under both aerobic and anaerobic conditions.
- The rate of energy utilization depends on several factors, such as the availability of oxygen, the demand for ATP, the hormonal regulation, and the feedback inhibition. The cell can adjust its energy utilization to match its energy needs and to maintain a balance between energy production and consumption.

Structure of mitochondria

- Mitochondria are small, rod-shaped organelles that are found in most eukaryotic cells, such as plants and animals .
- They are involved in cellular respiration, which is the process of converting organic molecules into energy (ATP) for the cell .
- They have a double membrane structure, consisting of an outer membrane and an inner membrane .
- The outer membrane is smooth and surrounds the entire organelle. It is permeable to small molecules and ions .
- The inner membrane is folded into projections called cristae, which increase the surface area for the reactions of cellular respiration .
- The space between the outer and inner membranes is called the intermembrane space, and the space enclosed by the inner membrane is called the matrix .
- The matrix contains various enzymes, molecules, and mitochondrial DNA, which is inherited maternally and encodes some of the proteins involved in cellular respiration .
- Mitochondria can divide by fission, which is similar to bacterial reproduction, and can fuse with other mitochondria to exchange genetic material and regulate their function .
- Mitochondria are thought to have evolved from ancient bacteria that were engulfed by ancestral eukaryotic cells and established a symbiotic relationship with them.
- Mitochondria are involved in many cellular processes, such as apoptosis (programmed cell death), calcium signaling, oxidative stress, and aging.
- Mitochondria can also be affected by various diseases, such as mitochondrial disorders, diabetes, cancer, neurodegenerative diseases, and cardiovascular diseases .

Cellular Respiration

- Cellular respiration is the process by which biological fuels are oxidised in the presence of an inorganic electron acceptor such as oxygen to produce large amounts of energy, to drive the bulk production of adenosine triphosphate (ATP).
- ATP is the main energy currency of the cell, and it is used to power various cellular processes such as biosynthesis, transport, and movement.
- Cellular respiration can be divided into four stages: glycolysis, pyruvate oxidation, the citric acid or Krebs cycle, and oxidative phosphorylation.

Structure of mitochondria

- Mitochondria are the organelles where most of the cellular respiration takes place in eukaryotic cells.

- Mitochondria have a double membrane structure: an outer membrane that surrounds the whole organelle, and an inner membrane that folds into cristae.
- The space between the two membranes is called the intermembrane space, and the space enclosed by the inner membrane is called the matrix.
- The inner membrane contains many proteins that are involved in the electron transport chain and oxidative phosphorylation, such as cytochromes, flavoproteins, and ATP synthase.
- The matrix contains enzymes that catalyse the reactions of pyruvate oxidation and the Krebs cycle, as well as mitochondrial DNA and ribosomes.

Glycolysis

- Glycolysis is the first stage of cellular respiration, and it occurs in the cytoplasm of the cell.
- Glycolysis is the breakdown of glucose, a six-carbon sugar, into two molecules of pyruvate, a three-carbon compound.
- Glycolysis involves ten enzymatic reactions, and it can be divided into two phases: the energy investment phase and the energy payoff phase.
- In the energy investment phase, two molecules of ATP are used to phosphorylate glucose and its intermediates, making them more reactive.
- In the energy payoff phase, four molecules of ATP and two molecules of NADH (a reduced form of nicotinamide adenine dinucleotide, an electron carrier) are produced for each molecule of glucose.
- The net gain of glycolysis is two molecules of ATP, two molecules of NADH, and two molecules of pyruvate per molecule of glucose.

Pyruvate oxidation

- Pyruvate oxidation is the second stage of cellular respiration, and it occurs in the mitochondrial matrix.
- Pyruvate oxidation is the conversion of pyruvate into acetyl coenzyme A (acetyl-CoA), a two-carbon compound that can enter the Krebs cycle.
- Pyruvate oxidation involves three steps: the decarboxylation of pyruvate, the oxidation of the remaining two-carbon fragment, and the transfer of the acetyl group to coenzyme A.
- Pyruvate oxidation is catalysed by a complex of enzymes called the pyruvate dehydrogenase complex.
- In pyruvate oxidation, one molecule of NADH and one molecule of carbon dioxide are produced for each molecule of pyruvate.

Krebs cycle

- The Krebs cycle, also known as the citric acid cycle, is the third stage of cellular respiration, and it occurs in the mitochondrial matrix.

- The Krebs cycle is a series of eight reactions that oxidise acetyl-CoA into carbon dioxide and generate energy carriers such as NADH, FADH₂ (a reduced form of flavin adenine dinucleotide, another electron carrier), and ATP.
- The Krebs cycle begins with the condensation of acetyl-CoA and oxaloacetate, a four-carbon compound, to form citrate, a six-carbon compound.
- The Krebs cycle then proceeds through a series of intermediates, such as isocitrate, alpha-ketoglutarate, succinyl-CoA, succinate, fumarate, malate, and oxaloacetate again.
- In each step of the Krebs cycle, a different enzyme catalyses the reaction and transfers electrons and protons to NAD⁺ or FAD, or phosphorylates ADP to ATP.
- The Krebs cycle produces two molecules

Relationship of carbohydrate metabolism to other compounds

- Carbohydrate metabolism is the process of breaking down and synthesizing carbohydrates, which are organic molecules composed of carbon, hydrogen, and oxygen atoms.
- Carbohydrates can be classified into simple sugars (monosaccharides and disaccharides) and complex sugars (polysaccharides), such as starch, glycogen, and cellulose.
- Carbohydrate metabolism is connected to the metabolism of other compounds, such as proteins and lipids, through common intermediates and pathways.
- The main pathways of carbohydrate metabolism are glycolysis, the formation of acetyl coenzyme A (acetyl CoA), the citric acid cycle (also known as the Krebs cycle), and the electron transport system and oxidative phosphorylation.
- Glycolysis is the first step of glucose catabolism, which is the breakdown of glucose to produce energy in the form of adenosine triphosphate (ATP). Glycolysis converts one molecule of glucose into two molecules of pyruvate, which can then enter the mitochondrial matrix for further oxidation.
- The formation of acetyl CoA is the link between glycolysis and the citric acid cycle. Pyruvate is oxidized and decarboxylated to form acetyl CoA, which is then transferred to the citric acid cycle.
- The citric acid cycle is a series of reactions that occur in the mitochondrial matrix and generate energy by oxidizing acetyl CoA to carbon dioxide and water. The citric acid cycle also produces reduced coenzymes, such as nicotinamide adenine dinucleotide (NADH) and flavin adenine dinucleotide (FADH₂), which carry electrons to the electron transport system.
- The electron transport system and oxidative phosphorylation are the final steps of glucose catabolism, which occur in the inner mitochondrial mem-

brane. The electron transport system consists of a series of protein complexes that transfer electrons from NADH and FADH₂ to oxygen, forming water and releasing energy. The energy is used to pump protons across the membrane, creating a proton gradient that drives the synthesis of ATP by the enzyme ATP synthase.

- Carbohydrate metabolism is connected to the metabolism of other compounds through the following ways :
 - The simple sugars that can also feed into glucose catabolism include galactose, fructose, glycogen, and pentose. These are catabolized during glycolysis or converted into glucose by various enzymes .
 - The amino acids from proteins can enter glucose catabolism through pyruvate, acetyl CoA, or components of the citric acid cycle, depending on their structure and degradation pathway . Some amino acids can also be synthesized from intermediates of glucose catabolism, such as oxaloacetate, alpha-ketoglutarate, and pyruvate.
 - The fatty acids from lipids can enter glucose catabolism through acetyl CoA, which is the product of beta-oxidation of fatty acids in the mitochondrial matrix . Fatty acids can also be synthesized from acetyl CoA by the process of lipogenesis in the cytosol.
 - The glycerol from lipids can enter glucose catabolism through glycolysis, after being converted into dihydroxyacetone phosphate by the enzyme glycerol kinase. Glycerol can also be synthesized from dihydroxyacetone phosphate by the reverse reaction.
 - The nucleotides from nucleic acids can enter glucose catabolism through the pentose phosphate pathway, which is an alternative pathway of glucose oxidation that produces ribose-5-phosphate, a precursor for nucleotide synthesis. Nucleotides can also be degraded into components that can enter glycolysis or the citric acid cycle, such as ribose, pyrimidines, and purines.

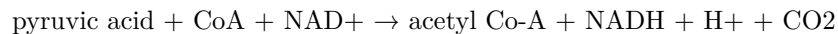
Glycolysis

- Glycolysis is a metabolic pathway that converts glucose, a six-carbon sugar, into two molecules of pyruvate, a three-carbon compound.
- Glycolysis occurs in the cytoplasm of the cell and does not require oxygen. It is the first stage of cellular respiration, which produces energy in the form of ATP and NADH from glucose.
- Glycolysis consists of ten enzyme-catalyzed reactions that can be divided into two phases: the preparatory phase and the payoff phase.
- In the preparatory phase, two molecules of ATP are used to phosphorylate glucose and split it into two molecules of glyceraldehyde-3-phosphate (G3P).
- In the payoff phase, each G3P molecule is oxidized and phosphorylated, producing two molecules of NADH and four molecules of ATP. The final product is two molecules of pyruvate.

- Glycolysis is an ancient metabolic pathway that is found in most living organisms. It can function in both aerobic and anaerobic conditions, depending on the availability of oxygen and the type of organism.
- In aerobic conditions, pyruvate can enter the mitochondria and undergo further oxidation in the Krebs cycle and the electron transport chain, producing more ATP and CO₂.
- In anaerobic conditions, pyruvate can be reduced to lactate or ethanol, depending on the organism, to regenerate NAD⁺ for glycolysis to continue. This process is called fermentation.

Formation of Acetyl Co-A

- Acetyl Co-A (acetyl coenzyme A) is a molecule that participates in many biochemical reactions in protein, carbohydrate and lipid metabolism.
- Its main function is to deliver the acetyl group to the citric acid cycle (Krebs cycle) to be oxidized for energy production.
- Acetyl Co-A formation occurs inside or outside the cell mitochondria. As a metabolite (a substance necessary for metabolism), acetyl Co-A must be freely available.
- It can be produced via the catabolism (breakdown) of carbohydrates (glucose) and lipids (fatty acids).
- Acetyl Co-A formation most commonly occurs during glucose catabolism. After carbohydrates have been broken down by digestive enzymes, the first stage of cellular glucose metabolism or glycolysis can begin. Glycolysis is the breaking down of glucose molecules.
- Glycolysis produces two molecules of pyruvic acid (pyruvate) per molecule of glucose. Pyruvate then enters the mitochondria, where it is converted into acetyl Co-A by a complex enzyme called pyruvate dehydrogenase.
- The overall formation reaction of acetyl Co-A from pyruvate may be represented as:



- This reaction may be called the oxidative decarboxylation of pyruvic acid to acetyl Co-A. It involves the removal of a carboxyl group (COOH) from pyruvate, releasing carbon dioxide, and the addition of a coenzyme A (CoA) to the remaining two-carbon fragment, forming acetyl Co-A.
- The reaction also transfers two electrons and a proton (H⁺) from pyruvate to the coenzyme NAD⁺, forming NADH. NADH is a high-energy electron carrier that can donate its electrons to the electron transport chain for ATP synthesis.

- Acetyl Co-A can also be formed from fatty acids through a process called beta-oxidation. Beta-oxidation occurs in the mitochondrial matrix and involves the sequential removal of two-carbon units from the fatty acid chain. Each two-carbon unit is then attached to CoA, forming acetyl Co-A.
- Acetyl Co-A is the common link between carbohydrate, lipid and protein metabolism. It can enter the citric acid cycle, where it is further oxidized and produces more NADH, FADH₂ and ATP. Alternatively, it can be used for biosynthesis of fatty acids, cholesterol, ketone bodies and other molecules.

Kreb cycle

The Kreb cycle, also known as the citric acid cycle or the tricarboxylic acid cycle, is a series of chemical reactions that occur in the mitochondria of aerobic organisms. It is the second stage of cellular respiration, following glycolysis and preceding the electron transport chain and oxidative phosphorylation. The Kreb cycle produces energy by oxidizing acetyl-CoA, which is derived from the breakdown of carbohydrates, fats, and proteins. The Kreb cycle also provides precursors for the synthesis of certain amino acids and other biomolecules, and generates reducing agents (NADH and FADH₂) that donate electrons to the electron transport chain.

The main steps of the Kreb cycle are:

- Acetyl-CoA (2 carbon atoms) combines with oxaloacetate (4 carbon atoms) to form citrate (6 carbon atoms) in a condensation reaction catalyzed by citrate synthase. This is the first and rate-limiting step of the cycle.
- Citrate is converted to isocitrate (6 carbon atoms) by a reversible isomerization reaction catalyzed by aconitase. This involves the removal and addition of a water molecule.
- Isocitrate is oxidized to alpha-ketoglutarate (5 carbon atoms) by a dehydrogenation reaction catalyzed by isocitrate dehydrogenase. This is the first oxidative decarboxylation reaction of the cycle, which releases a molecule of carbon dioxide and reduces a molecule of NAD⁺ to NADH.
- Alpha-ketoglutarate is further oxidized to succinyl-CoA (4 carbon atoms) by another oxidative decarboxylation reaction catalyzed by alpha-ketoglutarate dehydrogenase. This also releases a molecule of carbon dioxide and reduces a molecule of NAD⁺ to NADH.
- Succinyl-CoA is converted to succinate (4 carbon atoms) by a substrate-level phosphorylation reaction catalyzed by succinyl-CoA synthetase. This involves the transfer of a phosphate group from succinyl-CoA to GDP, forming GTP, which can then be converted to ATP by nucleoside diphosphate kinase. This is the only step of the cycle that directly produces

ATP (or GTP).

- Succinate is oxidized to fumarate (4 carbon atoms) by a dehydrogenation reaction catalyzed by succinate dehydrogenase. This is the only step of the cycle that occurs in the inner mitochondrial membrane, where succinate dehydrogenase is part of complex II of the electron transport chain. This reaction reduces a molecule of FAD to FADH₂, which can then donate electrons to the electron transport chain.
- Fumarate is converted to malate (4 carbon atoms) by a hydration reaction catalyzed by fumarase. This involves the addition of a water molecule across the double bond of fumarate.
- Malate is oxidized to oxaloacetate (4 carbon atoms) by a dehydrogenation reaction catalyzed by malate dehydrogenase. This reduces a molecule of NAD⁺ to NADH, which can then donate electrons to the electron transport chain.

The Krebs cycle completes one turn for each molecule of acetyl-CoA that enters it. Since each molecule of glucose produces two molecules of acetyl-CoA in the previous stage of glycolysis and pyruvate oxidation, the Krebs cycle turns twice for each molecule of glucose. The net products of the Krebs cycle per molecule of glucose are:

- 4 molecules of carbon dioxide (released as waste)
- 6 molecules of NADH (carrying high-energy electrons)
- 2 molecules of FADH₂ (carrying high-energy electrons)
- 2 molecules of ATP (or GTP) (providing immediate energy)

The Krebs cycle is regulated by several factors, such as the availability of substrates (acetyl-CoA, NAD⁺, FAD, ADP, etc.), the feedback inhibition by products (NADH, ATP, etc.), and the allosteric modulation by other metabolites (citrate, calcium, etc.). The Krebs cycle is also influenced by hormonal signals, such as insulin and glucagon, that affect the metabolism of carbohydrates, fats, and proteins. The Krebs cycle is essential for the energy production and biosynthesis of aerobic organisms, and its dysfunction can lead to various diseases, such as diabetes, cancer, and neurodegeneration.

Electron Transport System and Oxidative Phosphorylation

- Electron Transport System (ETS) is a series of proteins and electron carriers located in the inner mitochondrial membrane that transfers electrons donated by the reduced molecules NADH and FADH₂ through a series of electron acceptors, to oxygen.
- Oxidative Phosphorylation (OP) is a process involving a flow of electrons through the ETS, which creates a proton gradient across the inner mitochondrial membrane, and the use of this gradient to drive the synthesis of ATP by a protein complex called ATP synthase.

- The ETS consists of four main complexes: Complex I (NADH dehydrogenase), Complex II (succinate dehydrogenase), Complex III (cytochrome bc₁ complex), and Complex IV (cytochrome c oxidase). There is also a mobile electron carrier called coenzyme Q (or ubiquinone) that shuttles electrons between Complex I and II to Complex III, and another mobile electron carrier called cytochrome c that transfers electrons from Complex III to Complex IV.
- As electrons move through the ETS, they release energy that is used to pump protons from the mitochondrial matrix to the intermembrane space, creating an electrochemical gradient. This gradient is also called the proton-motive force, as it drives the protons back into the matrix through the ATP synthase, which uses the energy to phosphorylate ADP to ATP.
- The ETS and OP are coupled by the chemiosmotic theory, which states that the energy released by the electron transport is stored as a proton gradient, and that the energy of the proton gradient is used to drive ATP synthesis.
- The ETS and OP are the final stages of cellular respiration, which is the process of breaking down organic molecules (such as glucose) to release energy in the form of ATP. Cellular respiration consists of four main steps: glycolysis, formation of acetyl co-A, Krebs cycle, and ETS and OP.
- Glycolysis is the first step of cellular respiration, which occurs in the cytoplasm of the cell. It involves the splitting of glucose (a six-carbon sugar) into two molecules of pyruvate (a three-carbon compound), with the net production of 2 ATP and 2 NADH per glucose molecule.
- Formation of acetyl co-A is the second step of cellular respiration, which occurs in the mitochondrial matrix. It involves the oxidation of pyruvate to acetyl co-A (a two-carbon compound) by an enzyme complex called pyruvate dehydrogenase, with the release of CO₂ and the production of 1 NADH per pyruvate molecule.
- Krebs cycle (or citric acid cycle) is the third step of cellular respiration, which also occurs in the mitochondrial matrix. It involves the oxidation of acetyl co-A to CO₂ by a series of enzymes, with the production of 3 NADH, 1 FADH₂, and 1 ATP (or GTP) per acetyl co-A molecule.
- ETS and OP are the fourth and final step of cellular respiration, which occurs in the inner mitochondrial membrane. It involves the transfer of electrons from NADH and FADH₂ to oxygen by the ETS, with the generation of a proton gradient and the synthesis of ATP by OP.
- The total yield of ATP from cellular respiration depends on the efficiency of the ETS and OP, as well as the type of shuttle system that transports NADH from the cytoplasm to the mitochondria. In eukaryotic cells, there are two main shuttle systems: the malate-aspartate shuttle and the glycerol-3-phosphate shuttle. The malate-aspartate shuttle operates in the liver, heart, and kidney cells, and transfers NADH to the ETS as NADH, resulting in 3 ATP per NADH. The glycerol-3-phosphate shuttle operates in the muscle and brain cells, and transfers NADH to the ETS as FADH₂,

resulting in 2 ATP per NADH.

- Assuming the malate-aspartate shuttle is used, the total yield of ATP from cellular respiration is 38 ATP per glucose molecule (2 from glycolysis, 2 from Krebs cycle, and 34 from ETS and OP). However,

ATP and Cellular Respiration

- ATP (adenosine triphosphate) is the energy-carrying molecule used in cells because it can release energy very quickly.
- Energy is released from ATP when the end phosphate is removed, forming ADP (adenosine diphosphate) and a free phosphate.
- ATP can be regenerated from ADP and a phosphate by using energy from cellular respiration.
- Cellular respiration refers to the breakdown of glucose and other respiratory substrates to make energy carrying molecules called ATP.
- Cellular respiration is a metabolic pathway that uses glucose to produce ATP, which may be used as energy to power many reactions throughout the body.
- Cellular respiration is a series of chemical reactions that break down glucose and produce ATP, which are:
 - Glycolysis: the splitting of glucose into two molecules of pyruvate, producing 2 ATP and 2 NADH.
 - Pyruvate oxidation: the conversion of pyruvate into acetyl-CoA, producing 2 NADH and releasing 2 CO₂.
 - Citric acid cycle (or Krebs cycle): the oxidation of acetyl-CoA into 2 CO₂, producing 2 ATP, 6 NADH, and 2 FADH₂.
 - Oxidative phosphorylation: the transfer of electrons from NADH and FADH₂ to oxygen, producing 26-28 ATP and 6 H₂O.
- Cellular respiration is a catabolic process, meaning it breaks down molecules and releases energy.
- Cellular respiration can be aerobic (using oxygen as the final electron acceptor) or anaerobic (using other molecules as the final electron acceptor).
- Cellular respiration is influenced by several factors, such as temperature, pH, substrate availability, enzyme activity, and oxygen concentration.
- Cellular respiration is the main source of ATP for most living organisms, and it is essential for maintaining life and homeostasis.